Wave Height Distributions on Shallow Foreshores

This document describes a coastal engineering calculator for the local distribution of individual wave heights on shallow foreshores, based primarily on the **Composite Weibull Distribution** of Battjes and Groenendijk (2000).

The project computes characteristic individual-wave statistics from three local inputs:

- spectral significant wave height, H_{m0};
- local still-water depth, d;
- foreshore slope written as $1 : M$, with $\tan\alpha = 1/M$.

The project includes a native Windows C++ interface, a portable C++ command-line program, and implementations in Fortran, MATLAB, Jupyter/Python, HTML, and Android. Numerical methods and available diagnostics may differ between implementations.

The implementation includes the deep-water convergence treatment discussed by Caires and Van Gent (2012): the Battjes-Groenendijk result is not permitted to exceed the corresponding Rayleigh value, and the program switches directly to Rayleigh statistics when the normalized transitional wave height is sufficiently large.

Scope of the calculator

The calculator predicts the **local short-term distribution of individual zero-crossing wave heights** at a point on a shallow, gently sloping foreshore. It returns:

- free-surface variance m_0;
- root-mean-square individual wave height H_{rms};
- transitional wave height H_{tr};
- normalized transition $\widetilde{H}_{tr} = H_{tr}/H_{rms}$;
- Composite Weibull scale parameters H_1 and H_2 when the Battjes-Groenendijk branch is active;
- exceedance thresholds H_N and means of the highest 1/3, 1/10, 1/50, 1/100, 1/250, and 1/1000 fractions;
- uncapped CWD means, reference Rayleigh means, final means, and cap indicators;
- normalized means, ratios to $H_{1/3}$, and numerical diagnostics;
- the selected branch: `Composite Weibull (B&G)` or `Rayleigh`.

The program is a **point model**. It does not propagate a spectrum, solve wave transformation along a profile, calculate wave setup, calculate breaking dissipation, or resolve individual waves in time. The local H_{m0}, depth, and slope must already be known from measurements or from an appropriate wave-transformation model.

Historical development

Chronological overview

Period	Development	Relevance to this software
1880s	Rayleigh developed the probability distribution now bearing his name in the context of random vibrations and wave-like phenomena.	Provides the one-parameter reference distribution for linear random-wave heights.
1952	Longuet-Higgins connected Gaussian narrow-band sea-surface theory to the Rayleigh distribution of crest-to-trough wave heights.	Establishes the deep-water statistical baseline and the standard exceedance formulas.
1978	Battjes and Janssen formulated a random-wave breaking dissipation model based on an energy balance and bore-type dissipation.	Clarified that depth-induced breaking acts statistically on a population of random waves and selectively removes energy from the largest waves.
1979–1980	Goda and Longuet-Higgins documented finite-bandwidth and nonlinearity effects in measured wind-wave statistics.	Explains why the empirical deep-water value of $H_{rms}/\sqrt{m_0}$ used by Battjes and Groenendijk differs from the ideal narrow-band value.
1980s–1990s	Several shallow-water distributions were proposed, including transformed-Rayleigh, Glukhovskiy-type, modified Glukhovskiy, and Weibull formulations.	Demonstrated the need for depth-dependent upper-tail models, but single-shape distributions did not fully reproduce the observed change between low and high waves.
1998	Groenendijk's MSc thesis and the WL | Delft Hydraulics H3351 report developed and validated the Composed Weibull point model.	Established the two-branch distribution, its calibration database, the slope-dependent transition, and the calculation recipe.
2000	Battjes and Groenendijk published the Composite Weibull Distribution in *Coastal Engineering*.	Provides the principal scientific model implemented by this repository.
2012	Caires and Van Gent examined finite-depth and constant-depth behaviour and clarified deep-water convergence and flat-bottom limitations.	Motivates the Rayleigh switch, Rayleigh caps, and explicit warnings for constant-depth applications.

Period	Development	Relevance to this software
Present implementation	C++ GUI/CLI implement the same governing equations using a bracketed scalar solver.	Provides traceable wave statistics and a consistent engineering report.

From Gaussian surface elevation to individual wave heights

The historical transition from linear deep-water statistics to the Composite Weibull model is best understood as a sequence of increasingly realistic descriptions.

A linear, stationary, narrow-band random sea may be represented by a Gaussian surface-elevation process. Under those assumptions, the slowly varying envelope amplitude is Rayleigh-distributed, and the corresponding zero-crossing wave heights are also described by a Rayleigh law. This result is not merely an empirical curve fit: it follows from the statistics of two independent Gaussian quadrature components.

Finite spectral bandwidth, nonlinear crest-trough asymmetry, triad interactions, wave breaking, and dissipation progressively weaken that theoretical foundation. On a shallow foreshore, the principal distortion is not uniform across all wave heights. Smaller waves remain comparatively close to the Rayleigh population, whereas the largest waves are preferentially limited by breaking. The observed distribution therefore changes curvature in the upper tail. This is the central empirical observation behind the two-branch Composite Weibull representation.

Relation to random-wave transformation models

The calculator is intentionally narrower in scope than a spectral wave-transformation model. Historical models such as Battjes-Janssen calculate the spatial evolution of wave energy and breaking dissipation. The present software starts after that transformation problem has been solved: it accepts the **local** H_{m0}, local depth, and local slope and reconstructs the corresponding local distribution of individual wave heights.

The distinction is fundamental:

- a transformation model determines how m_0 changes along a profile;
- the Composite Weibull point model determines how the local energy is distributed among individual wave heights at one selected point;
- a structural, run-up, or overtopping method subsequently uses one or more of those local wave-height statistics.

The three stages should not be conflated.

Rayleigh distribution and early random-wave statistics

The Rayleigh distribution originated in nineteenth-century probability and vibration theory. Longuet-Higgins (1952) established its central role in ocean-wave statistics by showing that crest-to-trough wave heights in a narrow-banded linear Gaussian sea follow a Rayleigh distribution.

For deep-water or weakly depth-limited conditions, the Rayleigh cumulative distribution is

$$F_R(H) = 1 - \exp\left[-\left(H/H_{rms}\right)^2\right]$$

The corresponding exceedance probability is

$$P(H > h) = \exp\left[-\left(h/H_{rms}\right)^2\right]$$

This distribution is completely defined by one scale parameter. Consequently, all characteristic wave heights are related by fixed ratios.

Random-wave breaking models

Battjes and Janssen (1978) introduced a physically based energy-dissipation model for random waves breaking on a beach. Their work combined an energy balance, bore-type dissipation, and a depth-limited representation of the breaking-wave population. Although that model is not the Composite Weibull model implemented here, it established essential concepts used in later shallow-water random-wave modelling:

- depth-induced breaking selectively affects the largest waves;
- the local depth limits the upper part of the wave-height distribution;
- wave-energy dissipation and wave setup can be treated through averaged conservation equations;
- random breaking requires a statistical description rather than a single deterministic breaker height.

Modified shallow-water distributions

During the following decades, several alternatives to the Rayleigh distribution were proposed for finite and shallow depths, including Glukhovskiy-type, modified Glukhovskiy, Weibull, and transformed-Rayleigh models. Klopman (1996) developed a modified Glukhovskiy formulation intended to account for depth limitation using local wave energy and depth.

These single-form distributions improved some shallow-water predictions but did not reproduce the observed abrupt change in slope between the lower and upper portions of measured wave-height distributions on Rayleigh probability paper.

Groenendijk thesis and Delft Hydraulics report, 1998

Groenendijk's 1998 Delft University of Technology thesis, followed by the WL | Delft Hydraulics report H3351 by Groenendijk and Van Gent, developed the **Composed Weibull Distribution**. The terminology in the thesis and report is commonly written *Composed Weibull*; the later journal paper uses *Composite Weibull*. Both names refer to the same two-branch concept.

The model was calibrated and validated using laboratory data for plane foreshores with slopes from approximately $1:20$ to $1:250$. The principal observation was that measured distributions contain:

- a lower-wave region that remains approximately Rayleigh-shaped;
- an upper-wave region in which breaking suppresses the tail much more strongly;
- a distinct transitional height separating the two regions.

Battjes and Groenendijk, 2000

Battjes and Groenendijk (2000) published the model in *Coastal Engineering*. The paper fixed the Weibull exponents at

$$k_1 = 2.0$$

$$k_2 = 3.6$$

and parameterized the transitional height using local depth and foreshore slope. With a continuity condition and a second-moment constraint, the entire normalized distribution is controlled by the single parameter $\widetilde{H}_{tr}$.

Caires and Van Gent, 2012

Caires and Van Gent (2012) examined implementation details and the use of the Battjes-Groenendijk distribution in constant and finite depths. Two conclusions are directly relevant to this software:

1. When dimensional wave heights are recovered using the Battjes-Groenendijk H_{rms} parameterization, the calculated values can overshoot the Rayleigh limit before approaching an incorrect lower asymptote in deep water.
2. On shallow **flat bottoms**, which are outside the original calibration domain, the Battjes-Groenendijk distribution with a nominal $1:250$ slope underestimated measured high-wave quantiles by approximately 7% to 15% on average.

The present implementation therefore applies a Rayleigh switch and Rayleigh caps, and it explicitly treats constant-depth flat-bottom applications as outside the recommended model domain.

Physical basis

Why Rayleigh statistics change on a shallow foreshore

The Rayleigh model relies on an approximately linear, Gaussian, narrow-banded sea surface. These assumptions progressively deteriorate as waves propagate into shallow water.

Shoaling and nonlinear interactions. Interaction with the seabed increases nonlinearity. Triad interactions transfer energy between harmonics, producing sharper crests, flatter troughs, and non-Gaussian surface elevations.

Selective depth-induced breaking. The highest waves break first. Breaking removes energy preferentially from the upper tail of the individual-wave distribution, while smaller waves may continue to propagate with less modification.

Spatial adaptation. Breaking and statistical reshaping do not occur instantaneously at the depth where a breaker criterion is first reached. The wave field requires a finite propagation distance to adapt. This spatial lag explains why the foreshore slope appears in the transitional-height parameterization.

Two statistical populations

The Composite Weibull model represents the observed distribution using two Weibull branches:

- the first branch represents lower, substantially non-breaking waves;
- the second branch represents higher, breaking or recently broken waves.

The branches meet at H_{tr}, the transitional wave height.

Notation

Symbol	Meaning	Units
H	individual zero-crossing wave height	m
H_{m0}	spectral significant wave height, $4\sqrt{m_0}$	m
m_0	zero-order spectral moment or free-surface variance	m^2
H_{rms}	root-mean-square individual wave height	m
d	local still-water depth	m
M	denominator of a $1:M$ foreshore slope	-
$\tan\alpha$	local foreshore slope, $1/M$	-
H_{tr}	transitional wave height	m
H_1, H_2	Weibull scale parameters	m
k_1, k_2	Weibull shape exponents	-
$\widetilde{H}$	wave height normalized by H_{rms}	-
H_N	height exceeded with probability $1/N$	m
$H_{1/N}$	mean height of the highest $1/N$ fraction of waves	m
$\gamma(a,x)$	unnormalized lower incomplete gamma function	-
$\Gamma(a,x)$	unnormalized upper incomplete gamma function	-

A tilde denotes normalization with H_{rms}:

$$\widetilde{H} = H/H_{rms}$$

The notation H_N and $H_{1/N}$ must not be confused:

- H_N is an exceedance threshold;
- $H_{1/N}$ is the mean of all waves above that threshold.

The Windows C++ program reports both H_N and final $H_{1/N}$ for $N = 3, 10, 50, 100, 250, 1000$. Neither statistic represents a deterministic storm maximum.

Probability definitions and statistical conventions

Cumulative distribution, exceedance probability, and density

For a non-negative individual wave height H, the cumulative distribution function is

$$F_H(h) = P(H \le h), \qquad h \ge 0.$$

The survival or exceedance function is

$$Q_H(h) = P(H > h) = 1 - F_H(h).$$

Where the distribution is differentiable, the probability density is

$$f_H(h) = \frac{dF_H(h)}{dh}.$$

The density integrates to unity:

$$\int_0^\infty f_H(h)\, dh = 1.$$

Exceedance level H_N

The level H_N is exceeded, on average, once in every N individual waves. It is defined by

$$Q_H(H_N) = \frac{1}{N},$$

or equivalently,

$$F_H(H_N) = 1 - \frac{1}{N}.$$

H_N is a quantile or threshold. It is **not** the average of the highest waves.

Mean of the highest $1/N$ fraction, $H_{1/N}$

The mean of the highest $1/N$ fraction is the conditional mean above H_N:

$$H_{1/N} = E\left[H \mid H > H_N\right].$$

Because $P(H > H_N) = 1/N$,

$$H_{1/N} = N \int_{H_N}^\infty h\, f_H(h)\, dh.$$

This definition is used throughout the code. For example:

- $H_{1/3}$ is the arithmetic mean of the highest one-third of the individual waves;
- $H_{1/100}$ is the arithmetic mean of the highest one percent;
- $H_{1/1000}$ is the arithmetic mean of the highest 0.1 percent.

Neither H_N nor $H_{1/N}$ is a deterministic storm maximum. A storm maximum additionally depends on the number of waves, dependence between successive waves, sea-state duration, and nonstationarity.

Spectral and wave-by-wave quantities

The zero-order spectral moment is

$$m_0 = \int_0^\infty S_\eta(f)\, df,$$

where $S_\eta(f)$ is the variance-density spectrum of free-surface elevation. The spectral significant wave height is

$$H_{m0} = 4\sqrt{m_0}.$$

H_{m0} is obtained from the spectrum, while $H_{1/3}$ is obtained from an ordered sample or a probability distribution of individual zero-crossing wave heights. They are close under many deep-water conditions but are not identical by definition and need not remain close in shallow water.

Rayleigh distribution

Cumulative distribution and density

Using the Rayleigh scale $H_{rms,R} = H_{m0}/\sqrt{2}$, the cumulative distribution is

$$F_R(H) = 1 - \exp\left[-\left(H/H_{rms,R}\right)^2\right]$$

The corresponding density is

$$f_R(H) = \frac{2H}{H_{rms,R}^2} \exp\left[-\left(H/H_{rms,R}\right)^2\right]$$

Exceedance wave height

The wave height exceeded once, on average, in every N waves is obtained from $P(H > H_N) = 1/N$:

$$H_N = H_{rms,R}\sqrt{\ln N}$$

Mean of the highest fraction

For a Rayleigh distribution, the mean of the highest $1/N$ fraction is

$$H_{1/N} = N H_{rms,R} \Gamma\left(\frac{3}{2}, \ln N\right)$$

For the narrow-band Rayleigh relation $H_{rms,R} = \sqrt{8m_0}$ and $H_{m0} = 4\sqrt{m_0}$, the exact ratios used by the software are:

Statistic	Rayleigh ratio to H_{m0}
$H_{1/3}$	1.001075736951740
$H_{1/10}$	1.272734273369137
$H_{1/50}$	1.560113379974762
$H_{1/100}$	1.668233372358517
$H_{1/250}$	1.801017222497626
$H_{1/1000}$	1.984835590575388

These constants are used both in the direct Rayleigh branch and as upper caps on dimensional Battjes-Groenendijk results.

Rayleigh moments and scale relations

For a Rayleigh law of scale $H_{rms,R}$, the general moment is

$$E\left[H^r\right] = H_{rms,R}^r \Gamma\left(1 + \frac{r}{2}\right).$$

For $r = 2$,

$$E\left[H^2\right] = H_{rms,R}^2.$$

This Rayleigh root-mean-square scale is distinct from the empirical CWD input parameter H_{rms} used for diagnostics in both branches.

Under ideal narrow-band Gaussian theory,

$$H_{rms,R} = \sqrt{8m_0},$$

and therefore

$$H_{rms,R} = \frac{H_{m0}}{\sqrt{2}}.$$

The empirical Battjes-Groenendijk relationship used elsewhere in the calculator is intentionally different because it incorporates the observed finite-bandwidth dependence of individual wave heights.

Derivation of the Rayleigh mean-high-wave expression

The threshold exceeded once in N waves satisfies

$$\exp\left[-\left(\frac{H_N}{H_{rms,R}}\right)^2\right] = \frac{1}{N}.$$

Hence,

$$\left(\frac{H_N}{H_{rms,R}}\right)^2 = \ln N,$$

and

$$H_N = H_{rms,R}\sqrt{\ln N}.$$

Substitution into the conditional-mean integral gives

$$H_{1/N} = N\int_{H_N}^{\infty} h\,\frac{2h}{H_{rms,R}^2}\exp\left[-\left(\frac{h}{H_{rms,R}}\right)^2\right]dh.$$

Using the change of variable

$$u = \left(\frac{h}{H_{rms,R}}\right)^2,$$

the result is

$$H_{1/N} = NH_{rms,R}\Gamma\left(\frac{3}{2}, \ln N\right).$$

This exact expression is the source of the Rayleigh constants tabulated above.

Composite Weibull Distribution

Normalized variables

The governing equations are solved in terms of wave heights normalized by H_{rms}:

$$\widetilde{H} = \frac{H}{H_{rms}}, \qquad \widetilde{H}_1 = \frac{H_1}{H_{rms}}, \qquad \widetilde{H}_2 = \frac{H_2}{H_{rms}}, \qquad \widetilde{H}_{tr} = \frac{H_{tr}}{H_{rms}}.$$

Normalization removes the absolute dimensional scale from the distribution-shape problem. Once the normalized solution is known, dimensional values are recovered by multiplication by H_{rms}.

Survival function

The exceedance form is often more convenient than the CDF:

$$Q(H) = 1 - F(H).$$

For the Composite Weibull Distribution,

$$Q(H) = \begin{cases} \exp\left[-\left(\dfrac{H}{H_1}\right)^{k_1}\right], & H \le H_{tr}, \\ \exp\left[-\left(\dfrac{H}{H_2}\right)^{k_2}\right], & H > H_{tr}. \end{cases}$$

Continuity of the CDF automatically implies continuity of the exceedance probability at the transition.

Probability associated with the transition

Define

$$x_1 = \left(\frac{\widetilde{H}_{tr}}{\widetilde{H}_1}\right)^{k_1}, \qquad x_2 = \left(\frac{\widetilde{H}_{tr}}{\widetilde{H}_2}\right)^{k_2}.$$

The continuity condition requires $x_1 = x_2$. Denoting their common value by x_{tr},

$$x_{tr} = x_1 = x_2.$$

The exceedance probability at the transition is

$$P(H > H_{tr}) = \exp(-x_{tr}),$$

and the nominal recurrence count associated with the transition is

$$N_{tr} = \exp(x_{tr}).$$

For $N < N_{tr}$, the exceedance threshold lies below the transition and the mean-high-wave integral includes both branches. For $N \ge N_{tr}$, the threshold lies in the upper branch.

Scale-parameter relation implied by continuity

The two scale parameters are not independent. From

$$\left(\frac{H_{tr}}{H_1}\right)^{k_1} = \left(\frac{H_{tr}}{H_2}\right)^{k_2},$$

one may write

$$H_2 = H_{tr}\left(\frac{H_1}{H_{tr}}\right)^{k_1/k_2}.$$

The C++ implementation eliminates both scales in favour of a common transition exponent q. After finding q from the second moment, it recovers both scales and checks continuity.

General moment of the Composite Weibull Distribution

For any real moment order $r > -k_1$, the normalized piecewise moment can be expressed as

$$E\left[\widetilde{H}^r\right] = \widetilde{H}_1^r \gamma\left(1 + \frac{r}{k_1}, x_1\right) + \widetilde{H}_2^r \Gamma\left(1 + \frac{r}{k_2}, x_2\right).$$

The first term integrates the lower branch from zero to the transition. The second term integrates the upper branch from the transition to infinity.

For $r = 0$, the expression represents total probability and equals one when continuity is satisfied. For $r = 1$, it gives the normalized mean individual wave height. For $r = 2$, it gives the second moment used to impose the H_{rms} normalization.

Piecewise cumulative distribution

The dimensional cumulative distribution is

$$F(H) = \begin{cases} 1 - \exp\left[-\left(H/H_1\right)^{k_1}\right], & H \le H_{tr}, \\ 1 - \exp\left[-\left(H/H_2\right)^{k_2}\right], & H > H_{tr}. \end{cases}$$

The fixed shape exponents are

$$k_1 = 2.0$$

$$k_2 = 3.6$$

Because $k_1 = 2$, the lower branch has Rayleigh form. The larger exponent $k_2 = 3.6$ produces a more rapidly decaying upper tail.

Piecewise probability density

The corresponding density is

$$f(H) = \begin{cases} \frac{k_1}{H_1}\left(H/H_1\right)^{k_1 - 1} \exp\left[-\left(H/H_1\right)^{k_1}\right], & H \le H_{tr}, \\ \frac{k_2}{H_2}\left(H/H_2\right)^{k_2 - 1} \exp\left[-\left(H/H_2\right)^{k_2}\right], & H > H_{tr}. \end{cases}$$

The CDF is continuous at H_{tr}, but the density is generally discontinuous because $k_1 \neq k_2$. This is an accepted empirical simplification: the integral statistics required for engineering calculations remain well behaved.

Continuity condition

Continuity at the transition requires

$$F_1(H_{tr}) = F_2(H_{tr})$$

which is equivalent to

$$(H_{tr}/H_1)^{k_1} = (H_{tr}/H_2)^{k_2}$$

In normalized form,

$$\left(\widetilde{H}_{tr}/\widetilde{H}_1\right)^{k_1} = \left(\widetilde{H}_{tr}/\widetilde{H}_2\right)^{k_2}$$

Second-moment normalization

The normalized distribution must satisfy

$$E\left[\widetilde{H}^2\right] = 1$$

Defining

$$x_1 = \left(\widetilde{H}_{tr}/\widetilde{H}_1\right)^{k_1}$$

$$x_2 = \left(\widetilde{H}_{tr}/\widetilde{H}_2\right)^{k_2}$$

the second-moment condition is

$$\widetilde{H}_1^2\gamma\left(1+\frac{2}{k_1}, x_1\right) + \widetilde{H}_2^2\Gamma\left(1+\frac{2}{k_2}, x_2\right) = 1$$

The scalar solver enforces $x_1 = x_2 = q > 0$ and imposes the second moment directly. After calculating the scales, the program checks

$$R_m = \sqrt{\widetilde{H}_1^2\gamma\left(1+\frac{2}{k_1}, q\right) + \widetilde{H}_2^2\Gamma\left(1+\frac{2}{k_2}, q\right)} - 1,$$

and the continuity residual

$$R_c = \left(\widetilde{H}_{tr}/\widetilde{H}_1\right)^{k_1} - \left(\widetilde{H}_{tr}/\widetilde{H}_2\right)^{k_2}.$$

Local physical parameterization

Spectral variance

The input spectral significant wave height is converted to free-surface variance using

$$m_0 = (H_{m0}/4)^2$$

Root-mean-square individual wave height

The implementation uses the Battjes-Groenendijk empirical relation

$$H_{rms} = \left(2.69 + 3.24\frac{\sqrt{m_0}}{d}\right)\sqrt{m_0}$$

Equivalently,

$$\frac{H_{rms}}{\sqrt{m_0}} = 2.69 + 3.24\frac{\sqrt{m_0}}{d}$$

The coefficient 2.69 represents the broad-banded deep-water limit adopted by Battjes and Groenendijk following Goda's analysis of wind-wave data. It differs from the narrow-band Rayleigh relation $H_{rms}/\sqrt{m_0} = \sqrt{8} \approx 2.828427$.

This distinction is the reason dimensional Battjes-Groenendijk values require the deep-water convergence treatment described later.

Foreshore slope

For an input slope $1 : M$,

$$\tan\alpha = 1/M$$

The program expects the positive denominator M, not the tangent itself.

Transitional wave height

The dimensional transitional wave height is

$$H_{tr} = (0.35 + 5.8\tan\alpha)\, d$$

For a slope entered as $1 : M$,

$$H_{tr} = (0.35 + 5.8/M)\, d$$

The normalized transition controlling the CWD shape is

$$\widetilde{H}_{tr} = H_{tr}/H_{rms}$$

A relatively small $\widetilde{H}_{tr}$ places more of the distribution in the breaking-controlled upper branch. A large $\widetilde{H}_{tr}$ moves the transition into the far tail and causes the normalized CWD to approach Rayleigh statistics.

Degree of saturation

A useful dimensionless measure of local wave energy relative to depth is

$$\mu = \frac{\sqrt{m_0}}{d}.$$

Because $H_{m0} = 4\sqrt{m_0}$,

$$\mu = \frac{H_{m0}}{4d}.$$

The empirical H_{rms} relation can then be written as

$$\frac{H_{rms}}{d} = (2.69 + 3.24\mu)\,\mu.$$

Combining this expression with the transition-height formula gives a direct dimensionless expression for the governing transition:

$$\widetilde{H}_{tr} = \frac{0.35 + 5.8/M}{(2.69 + 3.24\mu)\,\mu}.$$

This equation shows explicitly that the normalized distribution shape is governed by two local dimensionless descriptors:

1. the energy-to-depth ratio μ;
2. the foreshore slope denominator M.

For fixed slope, increasing H_{m0}/d decreases $\widetilde{H}_{tr}$ and places more probability in the breaking-controlled upper branch. For fixed local energy and depth, a steeper foreshore increases H_{tr} and therefore increases $\widetilde{H}_{tr}$, reflecting the finite spatial distance required for the wave population to adapt to breaking.

Statistical wave-height calculations

Wave height exceeded once in every N waves

The candidate threshold from the lower branch is

$$\widetilde{H}_{N,1} = \widetilde{H}_1\,(\ln N)^{1/k_1}$$

The final exceedance height is

$$\widetilde{H}_N = \begin{cases} \widetilde{H}_1\,(\ln N)^{1/k_1}, & \widetilde{H}_{N,1} < \widetilde{H}_{tr}, \\ \widetilde{H}_2\,(\ln N)^{1/k_2}, & \widetilde{H}_{N,1} \geq \widetilde{H}_{tr}. \end{cases}$$

Mean of the highest $1/N$ fraction

When $\widetilde{H}_N < \widetilde{H}_{tr}$, the averaging integral crosses both Weibull branches:

$$\widetilde{H}_{1/N} = N\widetilde{H}_1 \left[\Gamma\left(1 + \frac{1}{k_1}, \ln N\right) - \Gamma\left(1 + \frac{1}{k_1}, x_1\right)\right] + N\widetilde{H}_2\Gamma\left(1 + \frac{1}{k_2}, x_2\right)$$

When $\widetilde{H}_N \geq \widetilde{H}_{tr}$, the averaging integral lies entirely in the upper branch:

$$\widetilde{H}_{1/N} = N\widetilde{H}_2\Gamma\left(1 + \frac{1}{k_2}, \ln N\right)$$

The dimensional result is

$$H_{1/N} = \widetilde{H}_{1/N} H_{rms}$$

The implementation evaluates these expressions for $N = 3$, 10, 50, 100, 250, and 1000. In the CWD branch, only the final *means* are limited by Rayleigh values; the thresholds remain those of the selected CWD.

Unified branch criterion

The lower-branch candidate threshold is

$$\widetilde{H}_{N,1} = \widetilde{H}_1 (\ln N)^{1/k_1}.$$

The same decision may be written using x_{tr}:

$$\ln N < x_{tr} \quad \Longleftrightarrow \quad \widetilde{H}_N < \widetilde{H}_{tr}.$$

Thus the branch is selected from a comparison between the requested exceedance level $\ln N$ and the transition exponent x_{tr}.

Derivation of the mean-high-wave integral

For a Weibull branch with scale H_s and exponent k,

$$f(H) = \frac{k}{H_s}\left(\frac{H}{H_s}\right)^{k-1} \exp\left[-\left(\frac{H}{H_s}\right)^k\right].$$

The substitution

$$u = \left(\frac{H}{H_s}\right)^k$$

gives

$$H = H_s u^{1/k}, \qquad dH = \frac{H_s}{k} u^{1/k-1} du.$$

Consequently,

$$\int H f(H)\, dH = H_s \int u^{1/k} e^{-u}\, du.$$

This integral is expressed by incomplete gamma functions. The two formulas used by the program follow by splitting the conditional-mean integral at H_{tr} whenever the exceedance threshold is below the transition.

Incomplete gamma functions

The complete gamma function is

$$\Gamma(a) = \int_0^\infty t^{a-1} e^{-t}\, dt$$

The unnormalized lower incomplete gamma function is

$$\gamma(a, x) = \int_0^x t^{a-1} e^{-t}\, dt$$

The unnormalized upper incomplete gamma function is

$$\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t}\, dt$$

They satisfy

$$\Gamma(a, x) = \Gamma(a) - \gamma(a, x)$$

The C++ solver evaluates incomplete gamma functions using a series for small arguments and a continued fraction for large arguments. It calculates the regularized upper tail directly, avoiding cancellation in rare-wave calculations. Its finite-interval gamma differences are evaluated by Gaussian quadrature. Other language implementations may use different numerical algorithms.

Normalized and unnormalized definitions

Some libraries return the regularized functions

$$P(a, x) = \frac{\gamma(a, x)}{\Gamma(a)}, \qquad Q(a, x) = \frac{\Gamma(a, x)}{\Gamma(a)}.$$

The governing equations in this repository use the **unnormalized** incomplete gamma functions. An implementation that substitutes P or Q without multiplying by $\Gamma(a)$ will produce incorrect moments and scale parameters.

Numerical evaluation strategy

The C++ implementation evaluates the regularized pair

$$P(a,x) = \gamma(a,x)/\Gamma(a), \qquad Q(a,x) = \Gamma(a,x)/\Gamma(a).$$

For $x < a + 1$, it evaluates P by a convergent power series; otherwise it evaluates Q by a continued fraction. The code converts these results to unnormalized functions when calculating moments. Both algorithms use an iteration limit of 10,000 and reject non-converged or non-finite results.

For upper-tail intervals whose endpoints are close, direct subtraction of two nearly equal gamma values is avoided. The C++ code uses a 16-point Gauss–Legendre rule to evaluate

$$\int_{u_1}^{u_2} u^{a-1} e^{-u}\, du,$$

and retains the upper incomplete gamma function for the infinite remainder. This protects the calculation of rare-wave mean heights from cancellation.

Nonlinear solution for $\widetilde{H}_1$ and $\widetilde{H}_2$

Scalar reduction of the governing equations

Let the normalized transition be

$$t = \widetilde{H}_{tr} = H_{tr}/H_{rms}.$$

By continuity, the two Weibull transition exponents are equal:

$$q = \left(\frac{t}{\widetilde{H}_1}\right)^{k_1} = \left(\frac{t}{\widetilde{H}_2}\right)^{k_2} > 0.$$

Therefore the normalized scales are reconstructed from a single scalar unknown:

$$\widetilde{H}_1 = tq^{-1/k_1}, \qquad \widetilde{H}_2 = tq^{-1/k_2}.$$

Substitution into the second-moment normalization gives

$$t^2 \left[q^{-2/k_1} \gamma\left(1 + \frac{2}{k_1}, q\right) + q^{-2/k_2} \Gamma\left(1 + \frac{2}{k_2}, q\right) \right] = 1.$$

With $k_1 = 2$ and $k_2 = 3.6$ this becomes

$$t^2 \left[\frac{\gamma(2,q)}{q} + q^{-5/9} \Gamma\left(\frac{14}{9}, q\right) \right] = 1.$$

This is the nonlinear scalar equation solved in the Windows C++ program.

Bracketed solution in logarithmic coordinates

The code uses $z = \ln q$ so that $q = e^z > 0$. It solves the logarithmic residual

$$R(z) = 2\ln t + \ln\left[q^{-2/k_1}\gamma\left(1+\frac{2}{k_1}, q\right) + q^{-2/k_2}\Gamma\left(1+\frac{2}{k_2}, q\right)\right] = 0.$$

The scalar residual is solved using a bracketed Brent method, combining bisection and interpolation. The solver checks the initial sign bracket, root convergence and reconstructed scale parameters. Invalid or non-finite states produce an explicit calculation error.

The solver constants are:

Parameter	Value
Regularized gamma relative tolerance	2.5×10^{-14}
Maximum gamma iterations	10,000
Maximum scalar iterations	200
Final logarithmic root residual	10^{-10}
Moment and continuity acceptance	10^{-10}

Verification of both CWD constraints

The scales are calculated as $\widetilde{H}_i = tq^{-1/k_i}$ and inserted into the original equations. The program independently verifies the second-moment residual

$$R_m = \sqrt{\widetilde{H}_1^2\gamma\left(1+\frac{2}{k_1}, q\right) + \widetilde{H}_2^2\Gamma\left(1+\frac{2}{k_2}, q\right)} - 1$$

and the continuity residual

$$R_c = \left(t/\widetilde{H}_1\right)^{k_1} - \left(t/\widetilde{H}_2\right)^{k_2}.$$

The solution is accepted only if both absolute residuals are below 10^{-10}. The report lists both residuals and the scalar iteration count; no empirical initial-scale guesses or finite-difference Jacobian are required.

Deep-water convergence and overshoot prevention

Origin of the dimensional inconsistency

The normalized Composite Weibull solution approaches the normalized Rayleigh distribution as $\widetilde{H}_{tr}$ becomes large. However, dimensional results are obtained by multiplying by the empirical Battjes-Groenendijk H_{rms}.

In the deep-water limit, that parameterization tends to

$$H_{rms} = 2.69\sqrt{m_0}$$

whereas the narrow-band Rayleigh scale used to derive the exact $H_{1/N}/H_{m0}$ limits is

$$H_{rms,R} = \sqrt{8m_0} \approx 2.828427\sqrt{m_0}$$

Caires and Van Gent showed that direct dimensionalization can therefore overshoot the Rayleigh prediction and later converge to a value below it. The program prevents this non-physical behavior with two safeguards.

Safeguard 1: Rayleigh branch

When

$$\widetilde{H}_{tr} > 2.75$$

the program bypasses the nonlinear Composite Weibull solution and directly assigns the exact Rayleigh $H_{1/N}/H_{m0}$ values listed above.

In this branch, H_1, H_2, and q are not calculated and appear as `N/A`. The Rayleigh reference uses $H_{rms,R} = H_{m0}/\sqrt{2}$, while the report also retains empirical H_{rms} as a separate diagnostic.

Safeguard 2: dimensional caps

When the Battjes-Groenendijk branch is active, every dimensional mean-high-wave statistic is capped as

$$H_{1/N} = \min\left(H_{1/N}^{BG}, C_N H_{m0}\right)$$

where C_N is the corresponding exact Rayleigh ratio.

After capping, the code calculates the final normalized value:

$$\widetilde{H}_{1/N}^{final} = H_{1/N}^{final}/H_{rms}.$$

The report distinguishes uncapped means, Rayleigh reference means and final means, with a `YES` or `NO` limit indicator. The exceedance threshold H_N is not capped.

Why a normalized Rayleigh limit is not sufficient

The CWD shape does approach Rayleigh as $\widetilde{H}_{tr} \to \infty$. That statement concerns normalized ratios based on the H_{rms} used inside the CWD. It does not by itself guarantee that dimensional values obtained from the empirical finite-bandwidth H_{rms} parameterization approach the exact Rayleigh values based on H_{m0}.

The correction is therefore applied in dimensional space. This distinction prevents an apparently converged normalized distribution from producing an inconsistent dimensional upper tail.

Behaviour at the switch

The threshold

$$\widetilde{H}_{tr} = 2.75$$

is an implementation criterion for deep-water convergence, not a physical breaker index. Immediately above the threshold, the direct Rayleigh constants are used. At or below the threshold, the CWD is solved and each final dimensional statistic is capped individually.

Because the limits apply separately to each conditional mean, a displayed final mean is not necessarily a moment of one modified probability distribution. The H_N thresholds always belong to the selected CWD or Rayleigh branch.

Calculation sequence

For each valid input set, the Windows C++ software:

1. Checks that H_{m0}, d, and M are finite and strictly positive.
2. Computes $m_0 = (H_{m0}/4)^2$, $\mu = \sqrt{m_0}/d$, and empirical H_{rms}.
3. Computes $\tan\alpha = 1/M$, H_{tr}, and $t = H_{tr}/H_{rms}$.
4. Uses Rayleigh directly when $t > 2.75$; otherwise solves the CWD scalar equation for q.
5. Reconstructs $\widetilde{H}_1$, $\widetilde{H}_2$ and verifies continuity and second moment.
6. Computes thresholds H_N and uncapped highest-fraction means for the six N values.
7. Calculates Rayleigh reference means $C_N H_{m0}$ and applies individual mean-height limits.
8. Recalculates normalized final means and ratios relative to $H_{1/3}$.
9. Reports diagnostics, branch identification, limit flags, and applicability warnings.
10. Displays the report and writes the same aligned text to `report.txt`.

Reference pseudocode

```
read Hm0, d, M; require finite positive values
sigma = Hm0 / 4
m0    = sigma^2
mu    = sigma / d
Hrms  = (2.69 + 3.24 * mu) * sigma
Htr   = (0.35 + 5.8 / M) * d
t     = Htr / Hrms

if t <= 2.75:
    solve scalar moment equation in z = log(q)
    H1tilde = t * q^(-1/2)
    H2tilde = t * q^(-1/3.6)
    check second-moment and continuity residuals
    branch = Composite Weibull (B&G)
else:
    branch = Rayleigh
    H1tilde = H2tilde = N/A

for N in {3, 10, 50, 100, 250, 1000}:
    H_N = exceedance threshold of selected branch
    raw_mean = upper-tail conditional mean
    rayleigh_mean = C_N * Hm0
    final_mean = min(raw_mean, rayleigh_mean) for CWD
```

```
    final_mean = rayleigh_mean for Rayleigh
    final_normalized = final_mean / Hrms

format report and write report.txt
```

Computational complexity

The calculation requires one scalar root solve only when the CWD branch is selected, followed by a small number of incomplete-gamma evaluations. Runtime is negligible relative to spectral wave propagation. Robustness of the special functions, positivity of the scalar root, and consistency of the reported statistics are more important than execution speed.

Inputs

H_{m0}: spectral significant wave height

Enter the local spectral significant wave height in metres. The value should correspond to the same location and water level as the input depth.

d: local water depth

Enter the positive local still-water depth in metres. The model does not add tides, storm surge, wave setup, or sea-level rise automatically.

M: slope denominator

Enter the denominator of the local foreshore slope $1 : M$.

Examples:

Foreshore slope	Input M	$\tan\alpha$
$1 : 20$	20	0.050000
$1 : 50$	50	0.020000
$1 : 100$	100	0.010000
$1 : 250$	250	0.004000

Use a representative local or averaged foreshore slope consistent with the point-model assumptions. Do not enter a slope angle in degrees.

Outputs

Distribution type

- `Composite Weibull (B&G)`: the scalar CWD equation was solved and both constraints were checked.
- `Rayleigh`: the normalized transition exceeded 2.75 and direct Rayleigh statistics were used.

Scale parameters

H_1 and H_2 are Weibull scale parameters, not breaker heights. Their normalized and dimensional values are reported in the CWD branch; they are marked `N/A` in the Rayleigh branch. The code also reports q, $P(H > H_{tr}) = e^{-q}$ and $N_{tr} = e^{q}$.

Mean-high-wave statistics

The program reports the six fractions 1/3, 1/10, 1/50, 1/100, 1/250, and 1/1000. For each it distinguishes:

- H_N: height exceeded with probability $1/N$ in the selected distribution;
- raw $H_{1/N}$: mean of the highest fraction before any cap;
- Rayleigh reference mean $C_N H_{m0}$;
- final $H_{1/N}$ and a `YES`/`NO` indicator of an applied limit.

The report also includes final $H_{1/N}/H_{rms}$ and never treats H_N as a storm maximum.

Diagnostic ratios

The report includes

$$H_{1/10}/H_{1/3}$$

$$H_{1/50}/H_{1/3}$$

$$H_{1/100}/H_{1/3}$$

$$H_{1/250}/H_{1/3}$$

$$H_{1/1000}/H_{1/3}$$

These ratios use the **final** mean-high-wave values and are dimensionless. The diagnostic section also reports the iteration count and the second-moment and continuity residuals for CWD cases.

Worked example

For

- $H_{m0} = 2.5$ m;
- $d = 5.0$ m;
- slope $1 : 100$;

then

$$m_0 = (2.5/4)^2 = 0.390625 \text{ m}^2$$

$$H_{rms} = 1.9344 \text{ m}$$

$$H_{tr} = (0.35 + 5.8/100) \times 5 = 2.0400 \text{ m}$$

$$\widetilde{H}_{tr} = 2.0400/1.9344 = 1.0546$$

Because $\widetilde{H}_{tr} < 2.75$, the `B&G` branch is used. The C++ implementation gives approximately:

Quantity	Result
$\widetilde{H}_1$	1.156695
$\widetilde{H}_2$	1.110155
Transition exponent q	0.831268
$H_{1/3}$ before limit	2.5911 m
Final $H_{1/3}$ (Rayleigh-limited)	2.5027 m
Final $H_{1/10}$	2.9701 m
Final $H_{1/50}$	3.3296 m
Final $H_{1/100}$	3.4567 m
Final $H_{1/250}$	3.6077 m
Final $H_{1/1000}$	3.8086 m

Here the raw $H_{1/3}$ exceeds its Rayleigh reference and is limited. The H_N exceedance thresholds, including $H_3 = 2.2043$ m, remain calculated from the CWD.

Rayleigh-branch example

For

- $H_{m0} = 2.5$ m;
- $d = 20.0$ m;
- slope $1 : 100$;

the intermediate values are

$$m_0 = \left(\frac{2.5}{4}\right)^2 = 0.390625 \text{ m}^2,$$

$$H_{rms} = \left(2.69 + 3.24\frac{0.625}{20}\right) 0.625 = 1.74453125 \text{ m},$$

$$H_{tr} = \left(0.35 + \frac{5.8}{100}\right) 20 = 8.16 \text{ m},$$

and

$$\widetilde{H}_{tr} = \frac{8.16}{1.74453125} \approx 4.6775.$$

Because $\widetilde{H}_{tr} > 2.75$, the direct Rayleigh branch is selected. The corresponding dimensional statistics are:

Quantity	Result
$H_{1/3}$	2.502689 m
$H_{1/10}$	3.181836 m
$H_{1/50}$	3.900283 m
$H_{1/100}$	4.170583 m
$H_{1/250}$	4.502543 m
$H_{1/1000}$	4.962089 m

The Rayleigh branch uses the scale $H_{m0}/\sqrt{2}$ for its exceedance thresholds and the tabulated coefficients for its conditional means. The empirical CWD H_{rms} remains in the intermediate results.

Sensitivity and dimensional interpretation

Dependence on H_{m0}/d

The degree of saturation is proportional to H_{m0}/d. Increasing local wave height at fixed depth, or reducing depth at fixed wave height, moves the distribution farther from Rayleigh. The largest changes generally occur in the high-N statistics because breaking primarily suppresses the upper tail.

Dependence on slope

At fixed H_{m0} and d, decreasing M means a steeper foreshore. The transition formula then gives a larger H_{tr} and a larger $\widetilde{H}_{tr}$. In the point-model parameterization, this leaves more of the distribution in the lower Rayleigh-shaped branch. The slope dependence should be interpreted as an empirical representation of spatial adaptation, not as a universal statement that steeper seabeds always produce larger physical waves.

Dimensional scaling

For fixed dimensionless conditions, the normalized quantities remain unchanged and dimensional wave heights scale with H_{rms}. However, changing H_{m0} at fixed depth also changes the degree of saturation, so practical scaling is not exactly linear unless the ratios H_{m0}/d and the slope are preserved.

Upper-tail sensitivity

As N increases, $H_{1/N}$ samples progressively rarer waves. These values are more sensitive to:

- the selected distribution;
- numerical precision in incomplete gamma functions;
- the deep-water cap;
- model-domain violations;
- measurement sampling uncertainty;

- the number of waves available for empirical validation.

Consequently, $H_{1/1000}$ should not be treated as equally well constrained as $H_{1/3}$ merely because both are produced by the same deterministic formula.

Model applicability

Original calibration domain

The underlying laboratory database included plane shallow foreshores with slopes approximately between

$$1 : 20 \text{ and } 1 : 250$$

The original studies identify an approximate domain that includes:

- degree of saturation approximately $0 < \sqrt{m_0}/d < 0.15$;
- equivalently, approximately $0 < H_{m0}/d < 0.6$ because $H_{m0} = 4\sqrt{m_0}$;
- straight, parallel depth contours;
- simple, gradually varying foreshore geometry;
- local wave conditions that have had sufficient propagation distance to adapt to the depth.

The source database included slopes $1 : 20$, $1 : 30$, $1 : 50$, $1 : 100$, and $1 : 250$ and predominantly JONSWAP-type laboratory spectra. The C++ output flags $M < 20$ or $M > 250$, and flags $\mu < 0.025$ or $\mu > 0.16$ when CWD is selected.

Point-model assumption

The distribution is assumed to depend primarily on the local m_0, d, and $\tan\alpha$, rather than on the full transformation history. This assumption is most defensible on a long, simple, monotonic foreshore.

Steep foreshores

For slopes steeper than approximately $1 : 20$, the transitional-height relation becomes less well supported and the spatial lag of breaking becomes increasingly important. A Rayleigh estimate may provide a conservative alternative, but project-specific data or a more suitable model should be preferred.

Very mild and flat bottoms

A flat bottom is not equivalent to a $1 : 250$ foreshore. On a constant-depth bed:

- waves are not continuously forced to break by a shoreward depth reduction;
- local wind input may be important;
- the wave-height distribution may remain closer to Rayleigh than the B&G model predicts;
- applying $M = 250$ can underestimate high-wave statistics.

Caires and Van Gent found mean underestimation of approximately 7% to 15% for high-wave quantities on shallow flat bottoms. For such cases, a Rayleigh distribution is generally the safer default unless a validated constant-depth model or site-specific measurements are available.

Bars, troughs, and rapidly varying bathymetry

The point model does not explicitly retain wave-field memory. On barred profiles, the distribution shoreward of a bar may gradually reform toward Rayleigh as waves enter deeper water, then depart from Rayleigh again if depth-induced breaking resumes. The simple local formula cannot fully represent this non-local evolution.

Use caution for:

- offshore bars and troughs;
- abrupt dredged channels;
- reef edges;
- submerged structures;
- rapidly changing slopes;
- strongly two-dimensional bathymetry;
- locations dominated by refraction, diffraction, or wave-current interaction.

Other limitations

The calculator does not explicitly include:

- wave period or spectral shape as an input;
- directional spreading;
- wave-current interaction;
- reflected waves;
- infragravity-wave separation;
- local wind growth;
- wave grouping;
- nonstationarity;
- individual-wave maximum statistics based on storm duration;
- uncertainty propagation.

The model should not be used outside its calibration domain without engineering judgement, sensitivity analysis, and comparison against alternative models or measurements.

Calibration evidence and interpretation of the stated range

The calibration range is not a rigid mathematical boundary at which the equations suddenly become invalid. It identifies the experimental region in which the parameters were fitted and checked. Predictions near the edge of the range should receive more scrutiny than predictions near the central portion of the database.

The original data represent controlled laboratory conditions. Field applications may include directional spreading, currents, irregular bathymetry, wind input, reflection, and mixed sea-swell systems that were not independently parameterized. Agreement of the input ratios with the laboratory range is therefore necessary but not sufficient evidence of applicability.

Recommended engineering checks

Before adopting a result, verify:

1. that H_{m0} and d refer to the same location and water level;
2. that the slope represents the adapting foreshore rather than an isolated local cell;

3. that the bathymetry is sufficiently monotonic for a point-model interpretation;
4. that the selected statistic is the one required by the downstream design method;
5. that Rayleigh, CWD, and any project-specific measured or modelled distributions have been compared;
6. that the result is not being interpreted as a deterministic maximum;
7. that sensitivity to water level, wave height, slope, and the 2.75 switch has been examined.

Engineering interpretation

Accurate upper-tail wave statistics are relevant to:

- armour stability and wave loading;
- revetment and dike design;
- run-up and overtopping assessments;
- crest-level studies;
- reliability analysis;
- interpretation of shallow-water physical-model data.

The calculator supplies wave-height statistics only. It does not replace the hydraulic formula, numerical model, physical model, design standard, or partial-safety framework required for the final engineering application.

For overtopping or structural calculations, verify which wave-height definition the selected design method requires. H_{m0}, $H_{1/3}$, an exceedance quantile, and the mean of the highest fraction are not interchangeable in shallow water.

Project files

File	Purpose
`shallow-water-waves_cli.cpp`	portable C++ command-line implementation
`shallow-water-waves_gui.cpp`	native Windows C++ graphical interface
`shallow-water-waves_cli.f90`	Fortran command-line implementation
`shallow_water_waves.m`	MATLAB function returning a results structure
`shallow-water-waves.ipynb`	Python/Jupyter implementation, tables, and plots
`weibull.apk`	Android application installation package
`manual.md`	theory, equations, limitations, build instructions, and usage

The Windows C++ GUI and portable C++ CLI share the scalar CWD formulation and aligned report. Other implementations should be checked individually for numerical details and available diagnostics.

C++ command-line program

Compile with GCC or MinGW-w64

Portable build:

```
g++ -O3 -std=c++17 -Wall -Wextra -pedantic \
    -Wconversion -Wsign-conversion \
    shallow-water-waves_cli.cpp \
    -o shallow-water-waves_cli
```

Optimized static Windows build, where the required static libraries are available:

```
g++ -O3 -march=native -std=c++17 -Wall -Wextra -pedantic \
    -Wconversion -Wsign-conversion \
    -static -static-libgcc -static-libstdc++ \
    shallow-water-waves_cli.cpp \
    -o shallow-water-waves_cli.exe
```

Run with command-line arguments

Linux or macOS:

```
./shallow-water-waves_cli 2.5 5.0 100
```

Windows:

```
shallow-water-waves_cli.exe 2.5 5.0 100
```

The argument order is:

```
Hm0  depth  slope_denominator
```

Run interactively

```
./shallow-water-waves_cli
```

When the three arguments are omitted, the program prompts for each input.

Output

The formatted report is printed to the terminal and written to

```
report.txt
```

in the current working directory.

Native Windows C++ GUI

Compile with MinGW-w64

```
g++ -O3 -march=native -std=c++17 -Wall -Wextra -pedantic \
    -Wconversion -Wsign-conversion -municode \
    shallow-water-waves_gui.cpp \
    -o shallow-water-waves_gui.exe \
    -mwindows -static -static-libgcc -static-libstdc++
```

Run

```
shallow-water-waves_gui.exe
```

Enter:

- `Hm0 (m)`;
- `Depth d (m)`;
- `Slope 1:M` with the denominator M;

then press **Compute**. The report is displayed in a monospaced output window and saved in UTF-8 as `report.txt`. Labels and numerical columns are aligned, and no extra blank lines follow the report.

Weibull Waves Android app

The Android app is written in Kotlin with Jetpack Compose and supports Android 6.0 (API 23) and newer.

It accepts the same three local inputs used by the other implementations: H_{m0}, water depth d, and slope denominator M for a slope $1 : M$. It reports the selected distribution branch, calculated parameters, dimensional and normalized mean-high-wave statistics, diagnostic ratios, and a copyable or shareable engineering report.

The `Weibull Theory` link opens an in-app explanation of the equations, numerical method, interpretation, and model limits.

Install `weibull.apk`

1. Copy `weibull.apk` to the phone or download it directly to the phone.
2. If requested, allow the browser or file manager opening the APK to install unknown apps.
3. Open `weibull.apk`, approve the installation, and launch `Weibull Waves`.
4. The install-from-source permission may be disabled again after installation.

With Android Platform Tools and USB debugging enabled, installation can also be performed with:

```
adb install -r weibull.apk
```

Fortran command-line program

Compile with GNU Fortran

```
gfortran -O3 -march=native -std=f2018 -Wall -Wextra -pedantic \
    -Wconversion -static -fno-underscoring \
    shallow-water-waves_cli.f90 \
    -o shallow-water-waves_cli_f
```

When static linking is unavailable, remove `-static`.

Run

```
./shallow-water-waves_cli_f 2.5 5.0 100
```

or interactively:

```
./shallow-water-waves_cli_f
```

The program prints the report and writes `report.txt`.

MATLAB function

Place `shallow_water_waves.m` on the MATLAB path or in the current working directory.

Return the result structure

```
results = shallow_water_waves(2.5, 5.0, 100);
```

Print the report without assigning an output

```
shallow_water_waves(2.5, 5.0, 100);
```

The function validates all three inputs, returns the full results structure, and writes `report.txt`.

Key structure fields include:

```
Hm0
d
slopeM
distribution_type
m0
Hrms
tanAlpha
Htr_dim
Htr_tilde
H1_Hrms
H2_Hrms
H1_3_Hrms ... H1_1000_Hrms
H1_dim
H2_dim
H1_3_dim ... H1_1000_dim
ratio_1_10_div_1_3 ... ratio_1_1000_div_1_3
```

Jupyter notebook

Create a virtual environment

Windows:

```
py -m venv .venv
.venv\Scripts\activate
python -m pip install --upgrade pip
pip install numpy scipy pandas matplotlib notebook
jupyter notebook shallow-water-waves.ipynb
```

Linux or macOS:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install numpy scipy pandas matplotlib notebook
jupyter notebook shallow-water-waves.ipynb
```

The notebook contains:

- theoretical background;
- incomplete-gamma functions;
- nonlinear solution with `fsolve`;
- the empirical initialization formulas;
- the main analysis workflow;
- formatted result tables;
- graphical analysis of the distribution and characteristic wave heights.

Run the cells sequentially and modify the input cell before executing the final analysis cells.

Numerical verification

Recommended cross-language verification steps:

1. Use identical H_{m0}, d, and M input in GUI and CLI.
2. Confirm the same distribution branch.
3. Compare m_0, H_{rms}, H_{tr}, and $\widetilde{H}_{tr}$.
4. For CWD, compare q, $\widetilde{H}_1$, $\widetilde{H}_2$, and both residuals.
5. Compare H_N, raw means, reference Rayleigh means, and final capped means.
6. Compare normalized means, limit indicators and final diagnostic ratios.
7. Confirm both residuals are below 10^{-10} for solved CWD cases.

Minor floating-point differences are expected, but engineering outputs should agree to the displayed precision.

Common interpretation errors

Confusing H_{m0} and $H_{1/3}$. They are approximately equal in deep water but not generally identical on a shallow foreshore.

Confusing H_N and $H_{1/N}$. One is a threshold of the selected distribution; the other is a conditional mean. Rayleigh limits apply to final conditional means, not to CWD thresholds.

Entering the tangent instead of the slope denominator. For $1 : 100$, enter `100`, not `0.01`.

Using the model on a flat bottom by entering an arbitrarily large M. The flat-bottom process is outside the original formulation and can produce non-conservative high-wave estimates.

Using offshore H_{m0} with local depth. The input H_{m0} must be local to the point being analysed.

Interpreting H_1 or H_2 as physical breaker heights. They are Weibull scale parameters.

Treating $H_{1/1000}$ as the maximum wave in a storm. Storm maximum statistics also depend on the number of waves and the duration of the sea state.

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